

CLAUDE.md

SECTION root TYPE document STATUS **active** OWNER Founder UPDATED 2026-08-26

Guidance for Claude Code when working in this repository.

What this is

ay2m/Office ("The Office") is the internal **documents repository** for Fly GACA — an independent, educational platform and open regulatory library for Saudi civil aviation. It holds every operating document that runs the company: strategy/OKRs, governance, legal, finance, KSA compliance, HR/people, operations/IT specs, GTM, customer success, investor relations, academy curriculum, and brand — **not** product code. A Claude Code session here is almost always **drafting, editing, or reorganizing organizational documents**, not writing or reviewing application code.

Product code lives in separate repos, all under the **ay2m** account:

Repo		What it is
ay2m/FlyGACA	private	The product. The bilingual web app (React 19 + Vite) <i>and</i> its Express backend on Cloud Run. The regulatory corpus and content pipelines live here too.
ay2m/Captain-Adel	private	The AI flight instructor service behind captadel.com
ay2m/FlyGACA-ios	public	The native SwiftUI family — one shared package, one App Store app per exam module
ay2m/FlyGACA-app	public, archived	The retired predecessor of ay2m/FlyGACA . Read-only, kept for its 1,005-commit history. Do not cite it as current.

iOS module status: **ELPT** and **AIP** ship. **PPL, CPL, IR and ATPL are parked** — their modules were removed from the iOS repo in 2026-08 pending a strategic decision, while their **web** study packs keep selling at the band prices in **03-finance/monetization.md**.

[!NOTE] Older documents in this tree refer to a **FlyGACA/...** org and to six per-module App Store metadata repos (**PPL** , **CPL** , **IR** , **ATPL** , **ELPT** , **AIP**). **Neither is current.** The **FlyGACA/...** paths are legacy redirects to **ay2m/...** , and no per-module repo exists under either owner — every one 404s, so App Store metadata lives in **ay2m/FlyGACA-ios** . Some documents also call **ay2m/FlyGACA** "the iOS family", or **ay2m/FlyGACA-app** "the web monorepo" — it is the web app and backend, and **FlyGACA-app** is archived.

[!IMPORTANT] Fly GACA is **not affiliated with GACA** (Saudi General Authority of Civil Aviation). This constraint is load-bearing across the whole document tree, not just user-facing product copy — legal, GTM, and investor materials must all state the relationship accurately.

This repo contains real operating material — signed/draft legal agreements, financial data, HR records, investor materials, strategic plans. Treat its contents as sensitive: don't summarize or quote its financial/legal/HR/investor content into other contexts (chat replies, other repos, external tools) beyond what the task actually requires, and never publish it. Security or data-isolation concerns go to the maintainer directly per `01-governance/SECURITY.md` — not a public issue.

Repository structure

Twelve numbered sections, each independently browsable:

#	Section	Contents
<code>00-strategy/</code>	Strategy	Annual plan & OKRs, master roadmap, CEO execution plan, Phase 0 tracker, numbered brainstorm
<code>01-governance/</code>	Governance	Founders' agreement, SHA, ESOP, Code of Conduct, board pack, decision log, <code>CLAUDE.md</code> (see below), <code>CONTRIBUTING.md</code> , <code>SECURITY.md</code>
<code>02-legal/</code>	Legal	NDA templates, EULA, SLA, pilot agreement, PDPL policies, DPA, IP/takedown procedure
<code>03-finance/</code>	Finance	Banking policy, procurement, expense policy, budget-vs-actual tracker, KPI dashboard, ZATCA invoicing (HTML tax-invoice + VAT-return templates)
<code>04-compliance-ksa/</code>	Compliance (KSA)	MISA license, ZATCA e-invoicing, PDPL DPIA, BCP/DR, Nitaqat plan
<code>05-people/</code>	People	Employment contracts, employee handbook, offer letters, onboarding/offboarding, HR policies
<code>06-operations-it/</code>	Operations/IT	Digital office setup, product specs (CRM, Captain Adel, Instructor Dashboard), hosting/secrets placement facts, <code>runbooks/</code> , <code>diagrams/</code> , <code>setup/</code>
<code>07-gtm/</code>	Go-To-Market	Sales playbook, demo script, objection handling, cold outreach, B2B pipeline, <code>seo/</code> strategy
<code>08-customer-success/</code>	Customer Success	Onboarding playbook, health scoring, NPS, QBR templates, at-risk/expansion playbooks
<code>09-investor-relations/</code>	Investor Relations	Pitch deck, FAQ, due-diligence questionnaire, investor update template, risk register
<code>10-academy-curriculum/</code>	Academy & Curriculum	Curriculum map, coverage matrix, PPL mock exams, B2C learner paths, instructor onboarding

#	Section	Contents
11-brand/	Brand	Design system, design tokens, style guide (Falcon Theme), logos, print assets (EN + RTL AR)

Support directories:

- `templates/` — reusable `.md` starters (`tpl-fin-report.md` , `tpl-hr-policy.md` , `tpl-legal-memo.md` , `tpl-ops-runbook.md` , `tpl-strat-proposal.md` ; mirrored under `ar/templates/`). Base new docs on these.
- `ar/` — a parallel Arabic (Saudi MSA) mirror of the **Markdown layer** across all 12 sections — same folder structure and filenames, translated content. It currently holds 119 `.md` against the English tree's 126, so **seven English docs have no Arabic counterpart yet** (the newest English additions run ahead of the mirror; four of the seven are in `04-compliance-ksa/`). The `.docx` / `.xlsx` / `.html` deliverables are EN-only. **English is authoritative** — on any conflict the English tree governs. Filenames stay ASCII kebab-case even under `ar/` for easy diffing.
- `tools/print/` — the Markdown → branded A4 PDF pipeline (see below), plus `check-facts.mjs` , the entity-facts gate described under **The family contract**.
- `contracts/flygaca-family.json` / `tools/contracts/` — the cross-repo family contract and the tool that stamps its self-hash. See **The family contract** below.
- `_print/` — generated PDFs (265 today), mirroring the whole tree (including `ar/`). Generated output, but is **committed** (not gitignored) — the CI gate below checks it's present and fresh.
- `_INDEX.md` / `ar/_INDEX.md` — master index of the whole tree.
- `ar/_GLOSSARY.md` — the EN↔AR terminology glossary that keeps translations consistent.
- `drive-index-updates.csv` — a root-level `old_path,new_path` map recording the Drive→repo filename normalisation (Title Case + spaces → ASCII kebab-case). Reference it when an old filename turns up in a link or an external index; it is data, not a doc, and the doc-check gate ignores it.
- `.claude/agents/` — four subagents scoped to this repo: **doc-smith** (any `.md` / `.html` add/edit/rename, front-matter, and the print pipeline — use it when docs-check fails), **ar-mirror** (translating a new/changed English doc into `ar/` against the glossary and rebuilding the Arabic PDFs), **ksa-compliance** (anything turning on Saudi regulation, and any vendored-skill output before it is adopted) and **family-warden** (the family contract, entity facts, and cross-repo drift). `agents/README.md` holds the roster and says when each applies; `06-operations-it/agent-workforce-plan.md` explains why the roster is four and not twelve, and how the family's other repos are meant to fit.
- `.claude/skills/` — six vendored Apache-2.0 governance/privacy/risk skills (GDPR controls, privacy impact assessment, ISO 27001, NIST 800-30 risk assessment, third-party vendor risk, PCI DSS) chosen because their output is written policy rather than shell commands. They are foreign-law scaffolding — **PDPL and ZATCA remain the governing regimes**, and `01-governance/` remains the source of policy. Provenance, the pinned upstream commit and the full guardrails: `.claude/skills/THIRD_PARTY_NOTICES.md` . Note the whole `.claude/` tree is in `SKIP_DIRS` for `check.mjs` / `build.mjs` / `build-html.mjs` , so nothing under it needs front-matter

or a `_print/` PDF — but that skip is root-anchored, so vendored skills must stay at `.claude/skills/`.

Several document formats coexist: polished deliverables are committed binaries (`.docx` / `.xlsx` / `.pptx` , plus source PDFs, investor-deck JPGs, and brand PSD/PNG assets — SVG logos/diagrams are committed as text — with `.gitattributes` declaring the binary set `.docx` / `.xlsx` / `.pptx` / `.pdf` / `.psd` / `.png` / `.jpg` / `.jpeg` / `.gsheet`); working notes, specs, drafts, and playbooks are `.md` — which is what the print pipeline and the doc-check CI both operate on. **20 pages are authored as HTML instead of Markdown** and render through `build-html.mjs` (below): the two bilingual investor decks in `09-investor-relations/decks/` , the four showcase pages (`00-strategy/brainstorms/00-strategic-brainstorms-dashboard.html` , `00-strategy/the-book-of-fly-gaca.html` , `11-brand/design-system.html` , `11-brand/tidal-reckoning.html`), the two ZATCA finance templates in `03-finance/` (`tax-invoice-template.html` , `vat-return-worksheet.html`), the eleven print-collateral sources in `11-brand/print/` (letterheads EN/AR — including the sealed Arabic sheet — memo, press release, business cards, envelope, compliments slip, contract cover; these share `11-brand/print/brand-print.css`), and `02-legal/authorization-letter-jawazat-2026-08-20.html` (the showcase/finance pages are self-contained). `tools/print/build-png.mjs` additionally re-screenshots the `11-brand/print/` sources into the 300 dpi PNGs the brand catalogue references.

The doc convention (enforced by CI)

Every content `.md` file requires YAML front-matter with these exact keys:

```
---
title: <doc title>
section: <e.g. 03-finance>
doc_type: <document|readme|template|...>
status: <active|draft|scaffold>
owner: <e.g. Founder>
last_updated: <YYYY-MM-DD>
lang: <en|ar>
---
```

Exemptions: the root `README.md` skips only the **front-matter** check — its PDF must still exist under `_print/` and stay fresh, like any other `.md` ; the checker never descends into `tools/**` or `_print/**` at all; files under any `templates/` directory still need a front-matter block but only the `title` key inside it (they use their own authoring schema, not the full doc metadata). Every other `.md` , anywhere in the tree (including `ar/`), needs the full set.

`status: draft` or `status: scaffold` stamps an automatic DRAFT/SCAFFOLD watermark on the rendered PDF — set it deliberately. (No doc currently uses `scaffold` ; the tooling supports it.)

Every `.md` must have a matching, up-to-date PDF under `_print/` . After creating or editing any `.md` file (English or Arabic), regenerate its PDF before committing:

```

cd tools/print
npm ci                                # once, or after a dependency change (no browser download –
uses the                               uses the
                                       # pre-installed Chromium the print pipeline auto-detects)
npm run build                         # incremental – only content-changed docs re-render
npm run build:force                   # rebuild every doc from scratch
node build.mjs <path/to/doc.md>      # render one file
node build-html.mjs                  # separately renders ALL 16 HTML pages to their _print/
PDFs
node build-png.mjs                   # re-screenshots 11-brand/print/*.html to the 300dpi
catalogue PNGs
node check.mjs                       # the CI gate itself – front-matter + Markdown-PDF
coverage +
                                       # staleness + HTML-PDF coverage; no browser needed, safe
anytime

```

(`tools/print/package.json` also exposes `npm run build:html` and `npm run check` aliases.) Adding or renaming **any** `.html` in the tree without running `build-html.mjs` fails CI, same as an unrebuilt `.md`.

`node check.mjs` is cheap, dependency-free and the single best thing to run before you commit — it reports exactly which docs are missing or stale (currently: 245 markdown + 20 brand-HTML docs in scope — it prints the live figures, so quote the command rather than copying these numbers). Forgetting the rebuild is the most common way this repo goes red; it has already happened once (four PDFs, fixed in `016f1b2`), so make the build part of the same commit as the doc edit rather than a follow-up.

`build.mjs` hashes each source file (folded with a hash of `theme.css` + `build.mjs` itself) into `tools/print/.buildcache.json`; `check.mjs` recomputes that hash and fails if it doesn't match the committed PDF, i.e. if you edited a doc but didn't rebuild its PDF. **Commit the regenerated PDF and the updated `.buildcache.json` together with the `.md` change** — a doc edit without a rebuilt PDF fails CI (`.github/workflows/docs-check.yml` — the only workflow — runs on push to `main` and on PRs touching `**/*.md`, `**/*.html`, `_print/**`, `tools/print/**`, or the workflow file itself; CI runs Node 20).

[!WARNING] **Editing `build.mjs` marks every PDF in the repo stale**, because its own bytes are folded into the shared `themeHash`. For a change that genuinely alters rendering, that's correct — rebuild. But for a change that *cannot* alter output (adding a `SKIP_DIRS` entry, a comment, a log line), re-rendering all 265 PDFs is pure churn: Chromium restamps `CreationDate` on every one, so all 265 show up as modified binaries. In that case re-stamp `.buildcache.json` with the new `themeHash` instead — recompute `sha256(themeHash + source)` per existing key and write it back with `JSON.stringify(cache, null, 1) + '\n'`. Verify with `node check.mjs` and a clean `git status _print/`.

Styling is the "Falcon Theme" documented in `11-brand/fly-gaca-document-style-guide.md`: Inter body · Cairo headings (and all Arabic text) · JetBrains Mono code, A4 page, 0.75in margins (0.9in bottom for the footer), footer page numbers, cover block generated from the front-matter. Fonts are vendored under `tools/print/fonts/` so the whole pipeline runs offline; it auto-detects

Chromium via `$PLAYWRIGHT_BROWSERS_PATH` (default `/opt/pw-browsers`) or `$CHROMIUM_PATH`. Requires Node 18+ and **Chromium $\geq$ 131** (CSS `@page` margin boxes drive the footer page numbers).

The family contract

`contracts/flygaca-family.json` is the one artifact the three active repos share. It is committed **byte-identically** to `ay2m/Office`, `ay2m/FlyGACA` and `ay2m/Captain-Adel`, and it exists because the family's cross-repo claims used to live only in prose and drifted without anything failing.

It holds three blocks, each naming the repo that **owns** it — only the owner edits its block, the other two copies are mirrors:

Block	Owner	Source of truth	Mirrored into
entity	this repo	01-governance/company-facts.md	ay2m/FlyGACA's <code>src/lib/seo/jsonld.ts</code> + both i18n bundles; ay2m/Captain-Adel's <code>footer.js</code> , <code>terms.html</code> , <code>privacy.html</code> , <code>package.json</code> , <code>LICENSE</code>
chat	ay2m/FlyGACA	server/src/contract.ts	the answer shape both brains must honour
repos	this repo	this file's repo table	the real roster — supersedes any prose citing a <code>FlyGACA/...</code> org

Each repo gates its own half in its existing CI: here it is `node tools/print/check-facts.mjs` (wired into `docs-check.yml`, aliased as `npm run check:facts`), which asserts every `entity` value against the `company-facts.md` table it was copied from — and asserts the IBAN and account number from that same doc are *absent* from the manifest, since the manifest travels to both product repos. The product repos run `tests/family-contract.test.ts` and `test/family-contract.test.js`.

To change it: edit the owning repo's copy, bump `version`, re-stamp the self-hash with `node tools/contracts/stamp-manifest.mjs contracts/flygaca-family.json`, copy the file verbatim into the other two repos, and open all three PRs together. Editing without re-stamping fails every repo's gate immediately.

Known limitation: nothing offline can prove the three copies are the same revision — `version` and `sha` only reduce that to a visible one-line diff. Closing it fully needs a scheduled cross-repo workflow, which does not exist yet.

Conventions

- **Bilingual, English-authoritative.** New or materially changed content docs should eventually get an `ar/` counterpart at the same path, translated per `ar/_GLOSSARY.md`'s terminology — but unlike the product repos, CI here does **not** hard-gate on EN/AR parity (only front-matter +

PDF freshness are checked independently per file), so an EN-only doc won't fail CI. Still keep the mirror in sync as a matter of practice; don't let it drift silently for long.

- **Sensitive-content discipline.** Legal, financial, HR, and investor documents in this repo are real operating material, not samples — quote or restate the minimum necessary for the task.
- **Governance is the source of policy**, not this file. `01-governance/CONTRIBUTING.md` is now a repo-native guide (rewritten 2026-08-09 — it used to be a vendored copy of the application repo's guide; its `npm run check:*` commands never applied here). `01-governance/SECURITY.md` is filled in and current — for security or data-isolation concerns email the maintainer directly (address in `SECURITY.md`). `01-governance/CODE_OF_CONDUCT.md` is real. `01-governance/company-facts.md` is the canonical entity-facts doc the *product* repos consume — check it before restating company facts anywhere.
- `01-governance/CLAUDE.md` used to be a stale vendored copy of a *different* repo's Claude guidance (the product monorepo's, describing `functions/`, routing, etc. — none of which exists here); it has been replaced with a short pointer back to this file. If you find it drifting again, fix the pointer, don't restore old content.
- License: Apache 2.0 (`01-governance/LICENSE`) for the repo's own material; regulatory content quoted anywhere belongs to GACA and is not covered by that license.

Where to look

📖 **Family context:** [The Blue of Fly GACA](#) — the founder's canon: the origin story, the tenets, and the voice the whole family is written in. It is a manifesto, not a spec. For the machine-readable side of the family — the repo roster, the legal-entity facts and the shared chat contract — see `contracts/flygaca-family.json`, which is committed byte-identically to all three active repos and gated by `tools/print/check-facts.mjs`.

- `_INDEX.md` — the readable master index across all 12 sections (and `ar/_INDEX.md` for the mirror). Note it defers to `00-strategy/00-master-office-paperwork-index.gsheel` as the authoritative master index.
- `00-strategy/roadmap.md` — current phase status and what's next.
- `06-operations-it/agent-workforce-plan.md` — the internal agent layer: what the four subagents are for, the earn-its-place test that caps the roster, the target rosters for `ay2m/FlyGACA` and `ay2m/Captain-Adel`, and what agents explicitly do **not** replace. Adopted as DEC-012.
- `tools/print/README.md` — full print-pipeline usage/config detail beyond the summary above.
- `01-governance/` — `company-facts.md`, `CODE_OF_CONDUCT.md`, `decision-log.md`, `CONTRIBUTING.md`, `SECURITY.md`.
- `ar/_GLOSSARY.md` — EN↔AR terminology, use it before translating anything.